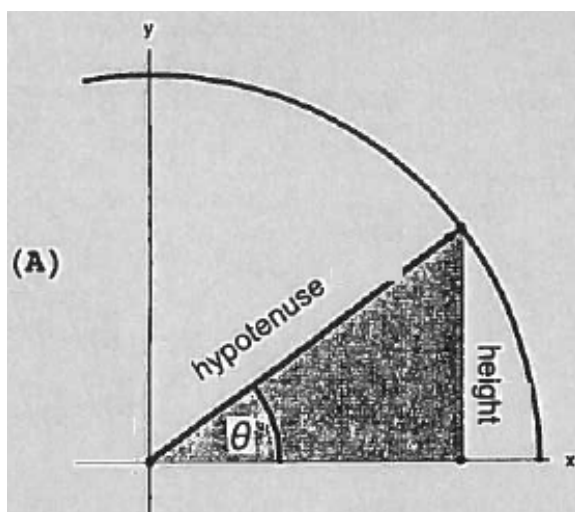


Problem 3

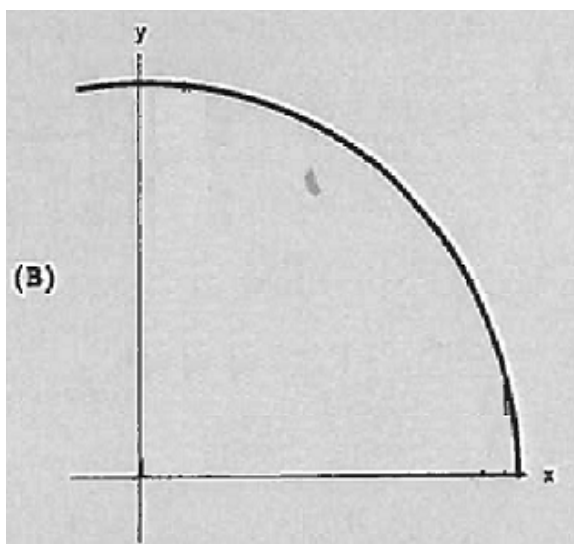
Problem 3

Problem 1 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see Figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis. Assume that the angle θ changes at the rate $\frac{d\theta}{dt} = 0.2$ radians per second.



- (a) Label Figure A.
- (b) In Figure B, draw the the triangle twice; once when θ is small and once more, when θ is close to $\frac{\pi}{2}$.

Problem 3



- (c) Find the rate of change of the height of the triangle when $\theta = \frac{\pi}{3}$
- (d) Find the rate of change of the area of the triangle when $\theta = \frac{\pi}{3}$